

Healthcare Insurance Risk Modelling Project Report

1. Executive Summary

This project develops a machine learning model to predict healthcare insurance charges using demographic and lifestyle information. The workflow covers exploratory data analysis, preprocessing, predictive modelling, model comparison, hyperparameter tuning and business interpretation. Four machine learning models were evaluated, with Gradient Boosting selected as the final solution because it achieved the highest predictive accuracy while maintaining strong generalization performance.

2. Project Objectives

- Predict insurance charges accurately.
- Identify the most influential pricing factors.
- Compare multiple machine learning algorithms.
- Generate actionable business recommendations for premium pricing and risk assessment.

3. Dataset & Methodology

The dataset contains customer demographic, lifestyle and health information including age, BMI, smoking status, gender, region and number of children. The target variable is insurance charges. The project follows an end-to-end workflow consisting of EDA, preprocessing using ColumnTransformer and One-Hot Encoding, train-test split, Pipeline implementation, model training, evaluation and interpretation.

4. Model Development

The following regression models were implemented and evaluated:

- Linear Regression – baseline model.
- Random Forest Regressor – ensemble tree model.
- Gradient Boosting Regressor – boosting-based ensemble.
- Hyperparameter Tuned Gradient Boosting using GridSearchCV.

5. Model Performance

Model	R ²	RMSE
Linear Regression	0.784	5796
Random Forest	0.864	4594
Gradient Boosting	0.879	4335
Tuned Gradient Boosting	0.878	4344

The original Gradient Boosting model slightly outperformed the tuned version on the unseen test data. Since hyperparameter tuning did not improve predictive performance, the original Gradient Boosting model was selected as the final production model.

6. Feature Importance

Gradient Boosting feature importance shows:

- Smoking Status – 67.7%
- BMI – 19.0%
- Age – 11.9%
- Children – <1%
- Gender & Region – negligible

These results indicate that health-related variables dominate insurance cost prediction while demographic variables contribute very little.

7. Business Insights

The analysis revealed several important business findings:

- Smoking is the strongest determinant of insurance charges and should be the primary pricing factor.
- Higher BMI and increasing age significantly increase expected insurance costs.
- Number of dependents has only a minor influence.
- Gender and geographic region have minimal predictive value and should not heavily influence premium pricing.
- Residual analysis confirmed that the selected model produces stable and unbiased predictions with no major systematic error patterns.

8. Business Recommendations

1. Introduce risk-based premium pricing centered on smoking status.
2. Include BMI and age as major premium adjustment variables.
3. Offer discounted premiums for healthy, non-smoking customers.
4. Avoid pricing discrimination based primarily on gender or location.
5. Deploy the Gradient Boosting model for automated premium estimation and underwriting support.

9. Conclusion

This project demonstrates a complete end-to-end machine learning workflow that combines data preprocessing, predictive modelling, model evaluation and business interpretation. The Gradient Boosting model achieved the best balance between predictive accuracy and generalization ($R^2 \approx 0.879$, $RMSE \approx 4335$). The results show that smoking status, BMI and age are the key drivers of insurance cost and provide practical guidance for premium calculation, customer risk segmentation and data-driven decision making.

